Kinwise.

Family medicine-review co-pilot · private, on-device AI

UnivaBio 2026 · AI for Human Health · Fictional demo data

Working prototype · web app, no backend

Helps families caring for an older adult on many medicines notice possible side effects early, and bring a clear, evidence-backed question to the doctor or pharmacist.

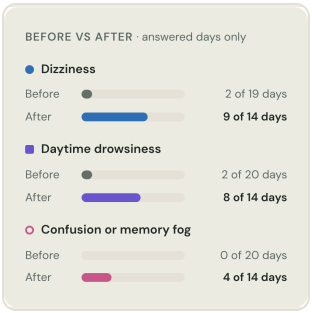
THE PROBLEM

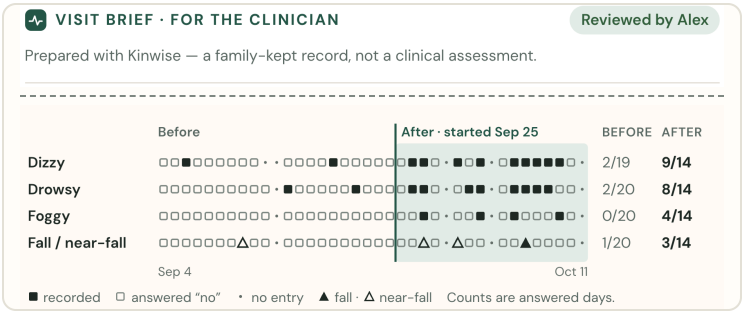
43.0%
of US adults 65+ used **5 or more prescription drugs** in a 30-day period (2017–Mar 2020), up from 33.3% in 2001–04.
NCHS, Health, United States

12.1 per 1,000 a year
emergency visits for medication harms, adults 65+ (2017–2019), vs 5.0 under 65; **95.8%** with medicines taken as intended.
Budnitz et al., JAMA 2021

More than 1 in 4
older adults **fall each year**, but **less than half tell their doctor**.
CDC, Facts About Older Adult Falls

Clinicians have the criteria but not the home observations; diary apps, the reverse. Kinwise connects the two for the visit.





Left: the caregiver Overview's "Before vs after" card for lorazepam (answered days only).

Right: the Visit brief after Alex ticks "I've reviewed this brief": its header and the day-by-day strip from section 2.

Screenshots of the live app. The full printout is attached: **kinwise-sample-visit-brief.pdf**.

WHO IT IS FOR

The **family caregiver** (Alex in the demo) and the **older adult** (Margaret, 78, 8 medicines from several prescribers), who checks in on a large-text screen. Written for her **doctor or pharmacist**.

WHAT IT DOES

1. **Check in, in her own words.** On-device NLP turns "felt woozy… nearly went over in the kitchen" into diary answers she confirms, and skips "my granddaughter fell" (ConText).
2. **Keep the medicine list.** Brands, combinations ("Tylenol PM") and misspellings match to 427 medicines.
3. **Review against published criteria.** AGS Beers 2023, anticholinergic burden (ACB 7), fall-risk lists, cascades; each finding is cited and ends as a question.
4. **See what changed.** Dizziness on **2 of 19** answered days before lorazepam started, **9 of 14** after (Fisher's exact test).
5. **Bring one page to the visit.** A brief the family reviews before printing, plus a FHIR R4 export.

PRIVACY & SAFETY BY DESIGN

No account, no server: the record stays in the browser and the model runs on the device (only model files are downloaded); nothing leaves unless the family prints or exports. Kinwise is designed never to say a medicine caused a symptom or to advise stopping one; screens carry *"Don't stop or change any medicine without talking to the prescriber."* No flag does not mean a medicine is free of risk.

HOW IT WORKS

Cited KB → Engines decide → Evidence ids → LLM writes → Code checks → Person reviews

Deterministic TypeScript engines over a cited knowledge base (427 medicines, 89 Beers rules, 26 cascades) compute every fact as a sentence with an id. The optional on-device LLM (Qwen2.5 on WebGPU) only rewords cited facts; code checks numbers, medicine names and wording and drops failing sentences. Without WebGPU, the template brief works the same way.

EVIDENCE & RIGOR

- **Clinical NLP:** micro F1 **0.942** (precision 0.972, recall 0.914) on 70 held-out diary stories never used for tuning; dev set (245 stories): 0.998.
- **Medication review: 57 of 57** synthetic cases and 295 of 295 expectations pass.
- **FHIR:** sample Bundle (270 resources): **0 errors** on the public HAPI R4 validator (25 warnings: unknown code systems).
- **On-device AI** (Qwen2.5-3B-Instruct (q4f16), Apple M1 Pro): 0 external requests; answers 4–12 s, brief 57.6 s. A live run found 3 AI outputs the checker should have stopped (a garbled sentence, a "can cause" explanation, a mis-quoted diary suggestion), so the template stays the default.
- **Software:** 1,245 unit tests pass (1 skipped); 20 of 20 browser end-to-end checks pass; 46 axe accessibility scans, 0 violations reported.

WORKING NOW (IN THE PROTOTYPE)

- Check-in with on-device NLP · medicine name matching
- Cited medication review · before/after statistics · record Q&A
- Visit brief, print and FHIR R4 export · optional on-device LLM

PLANNED (NOT BUILT)

- Clinician review of the KB (not yet done); user study
- Import from pharmacy or portal · family sharing · Spanish
- Public hosted demo (GitHub Pages)

Code: github.com/albertzorzolan/UnivaBioBabson (branch medwatch-v2)
Live demo (planned, not yet deployed): albertzorzolan.github.io/UnivaBioBabson/
Team: [Team member names, roles]

Hackathon prototype: not reviewed or cleared by the FDA (no exemption claimed), not clinically validated, no HIPAA claim. The doctor or pharmacist decides. In an emergency, call 911.